

Om Namah Shivay...



AMRITA
VISHWA VIDYAPEETHAM

School of Engineering
Chennai Campus

23MEE203_Metallurgy and Materials Science



Why study materials science?

Classification of materials:

Metals

Metals are composed of one or more metallic elements (e.g., iron, aluminum, copper, titanium, gold, nickel), and often also nonmetallic elements (e.g., carbon, nitrogen, oxygen) in relatively small amounts.

Ceramics

Ceramics are compounds between metallic and nonmetallic elements; they are most frequently oxides, nitrides, and carbides. For example, common ceramic materials include aluminum oxide (or alumina, Al_2O_3), silicon dioxide (or silica, SiO_2), silicon carbide (SiC), silicon nitride (Si_3N_4), and, in addition, what some refer to as the traditional ceramics—those composed of clay minerals (e.g., porcelain), as well as cement and glass.

Polymers

Polymers include the familiar plastic and rubber materials.



Composites

A composite is composed of two (or more) individual materials that come from the categories previously discussed—metals, ceramics, and polymers.

Advanced Materials:

Materials utilized in high-technology (or high-tech) applications are sometimes termed *advanced materials*.

Semiconductors

Semiconductors have electrical properties that are intermediate between those of electrical conductors (i.e., metals and metal alloys) and insulators (i.e., ceramics and polymers).

Biomaterials

Replacement implants are constructed of biomaterials—nonviable (i.e., nonliving) materials that are implanted into the body, so that they function in a reliable, safe, and physiologically satisfactory manner, while interacting with living tissue.



23MEE203_Metallurgy and Materials Science

Smart Materials

Smart (or intelligent) materials are a group of new and state-of-the-art materials now being developed that will have a significant influence on many of our technologies.

Nanomaterials

One new material class that has fascinating properties and tremendous technological promise is the nanomaterials, which may be any one of the four basic types—metals, ceramics, polymers, or composites. However, unlike these other materials, they are not distinguished on the basis of their chemistry but rather their size; the nano prefix denotes that the dimensions of these structural entities are on the order of a nanometer (10^{-9} m)—as a rule, less than 100 nanometers (nm)—which is equivalent to approximately 500 atoms.



23MEE203_Metallurgy and Materials Science

Crystallography

It is a branch of science in which internal structure of crystal, their properties, external and internal symmetries of crystal are studied along with the study of geometric form and other physical properties. It is useful for the identification of the minerals. The later are chemical substances formed under natural conditions and have crystal forms.

Solids (crystals or solids):

Solid materials are classified according to the regularity with which atoms are arranged with respect to one another and are classified as crystalline or non-crystalline (amorphous) materials or solids.

Crystalline solids- If the atoms of a solid are arranged in a pattern that repeats itself in three dimensions, they form a solid that has long range order (LRO) and are referred as crystalline solid. Examples are metals, alloys, some ceramic materials etc. A crystalline material may be either in the form of single crystal or an aggregate of many crystals known as polycrystalline.

Non crystalline or amorphous solids- Materials or solids whose atoms and ions are not arranged in a long range, periodic and repeatable manner, possess only short-range order and exist only in immediate neighbourhood of an atom or a molecule are called non crystalline or amorphous solids. Examples are most polymers, ceramic, glasses, and some metals.



23MEE203_Metallurgy and Materials Science

Structure- The arrangement and disposition of atoms within a crystal is called structure.

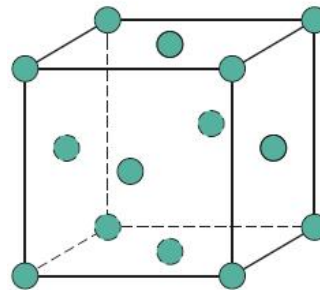
Crystal- The systematic geometric pattern arrangement of atoms and molecules in a solid is called crystal.

Space lattice- A space lattice is an array of points showing how particles (atoms, ions or molecules) are arranged at different sites in three dimensional spaces. It is a periodic arrangement in regular manner and having repeat distances a , b and c (vectors) in three directions called fundamental lattice vectors. If atoms are placed at these lattice positions, we obtain a crystalline solid.

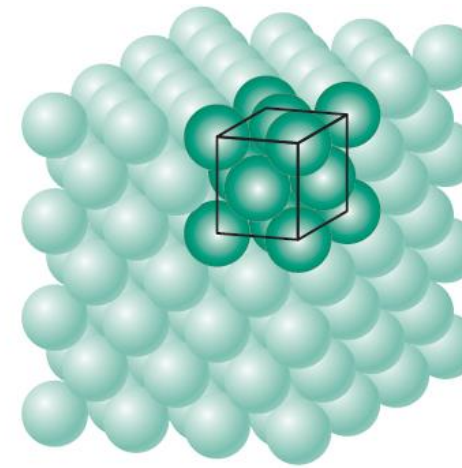
Unit cell

The smallest repeating unit of the crystal lattice is the unit cell, the building block of a crystal. A unit cell is the basic structural unit or building block of the crystal structure and defines the crystal structure by virtue of geometry and atom position.

Figure 1 (a) a reduced sphere unit cell, and (b) an aggregate of many atoms.



(a)



(b)



23MEE203_Metallurgy and Materials Science

Types of Unit Cell:

Numerous unit cells together make a crystal lattice. Constituent particles like atoms, molecules are also present. Each lattice point is occupied by one such particle.

- Primitive Cubic Unit Cell
- Body-centered Cubic Unit Cell
- Face centered cubic unit cell

The number of atoms per unit cell, N , can be computed using the following formula:

$$N = N_i + \frac{N_f}{2} + \frac{N_c}{8}$$

where

N_i = the number of interior atoms

N_f = the number of face atoms

N_c = the number of corner atoms

Number of atoms in **simple cubic or SC**: In each cubic unit cell, there are 8 atoms at the corners. Therefore, the total number of atoms in one unit cell is

$$N = N_i + \frac{N_f}{2} + \frac{N_c}{8} = 0 + \frac{0}{2} + \frac{8}{8} = 1 \text{ atom}$$

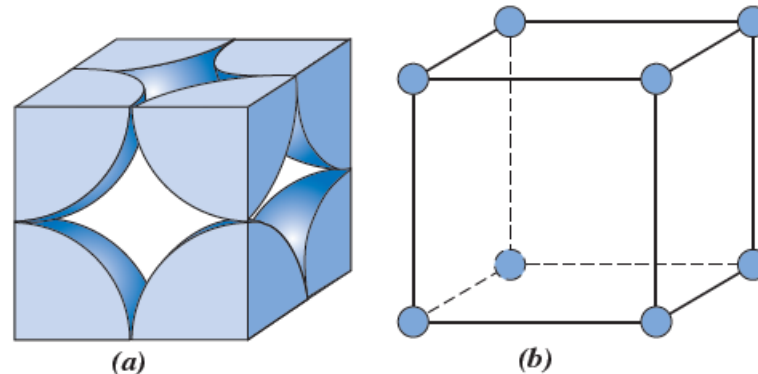


Figure 2 simple cubic unit cell



23MEE203_Metallurgy and Materials Science

Number of atoms in **BCC Cell**:

$$N = N_i + \frac{N_f}{2} + \frac{N_c}{8}$$

$$N = 1 + \frac{0}{2} + \frac{8}{8} = 2 \text{ atoms}$$

Number of atoms in **FCC Cell**:

$$N = N_i + \frac{N_f}{2} + \frac{N_c}{8}$$

$$N = 0 + \frac{6}{2} + \frac{8}{8} = 4 \text{ atoms}$$

Number of atoms in **HCP (Hexagonal closed pack) Cell**:

$$N = N_i + \frac{N_f}{2} + \frac{N_c}{6} = 3 + \frac{2}{2} + \frac{12}{6} = 6 \text{ atoms}$$

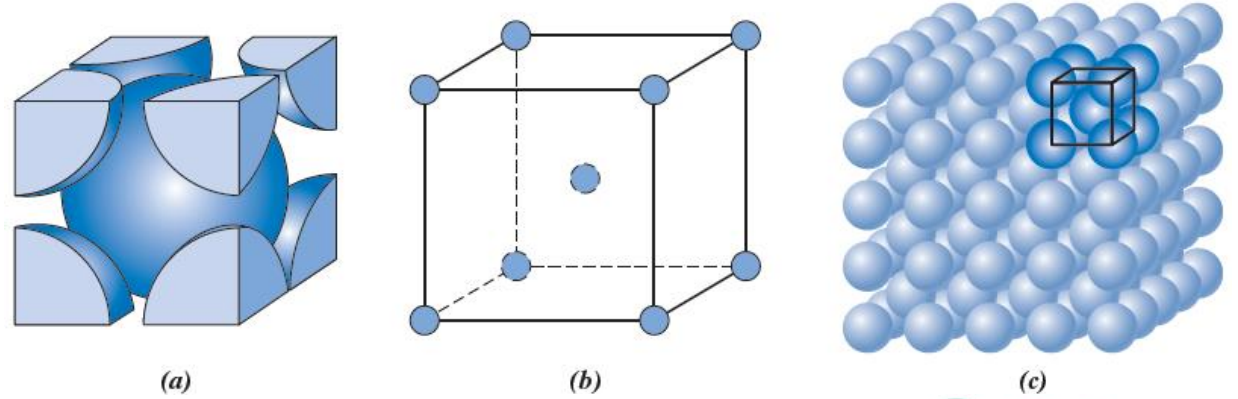


Figure 3 BCC unit cell

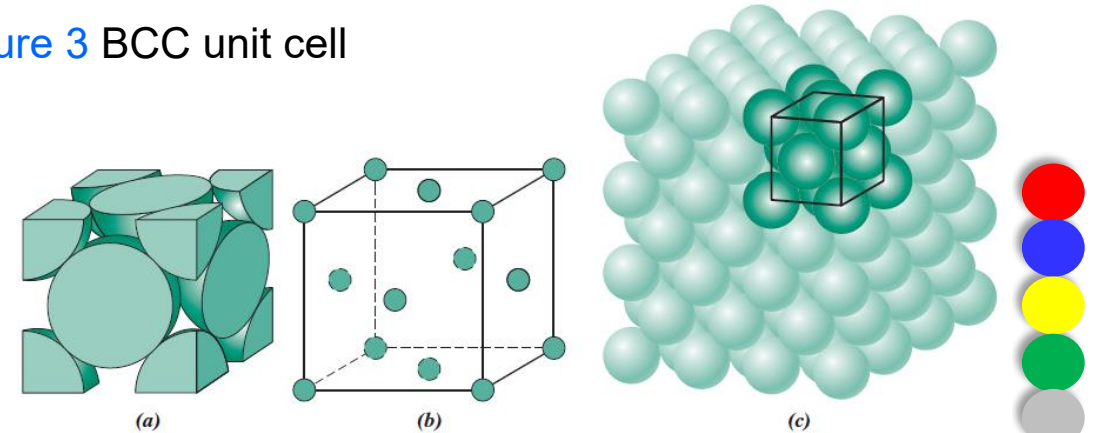


Figure 4 FCC unit cell

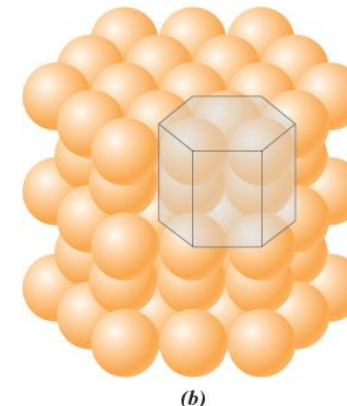
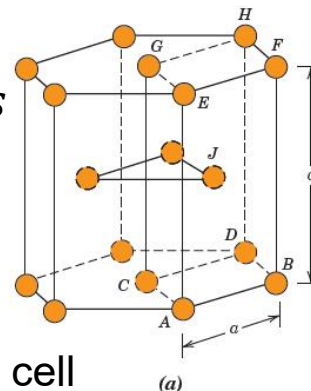


Figure 5 HCP unit cell

23MEE203_Metallurgy and Materials Science

Two other important characteristics of a crystal structure are the **coordination number** and the **atomic packing factor (APF)**. Each atom has the same number of nearest-neighbor or touching atoms for metals, which is the **coordination number**. For face-centered cubic, the coordination number is 12. This may be confirmed by examining Figure 4a: the front face atom has four corner nearest-neighbor atoms surrounding it, four face atoms in contact from behind, and four other equivalent face atoms residing in the next unit cell to the front (not shown).

Atomic packing factor (APF):

It can be defined as the ratio between the volume of the basic atoms of the unit cell (which represents the volume of all atoms in one unit cell) to the volume of the unit cell itself. For cubic crystals, A.P. F depends on the radius of atoms and characterization of chemical bonding.

$$APF = \frac{\text{Volume of atoms in a unit cell}}{\text{Total unit cell volume}} = \frac{N_{atoms} \times V_{atom}}{V_{unit\ cell}}$$



In general, less CN less is the APF



23MEE203_Metallurgy and Materials Science

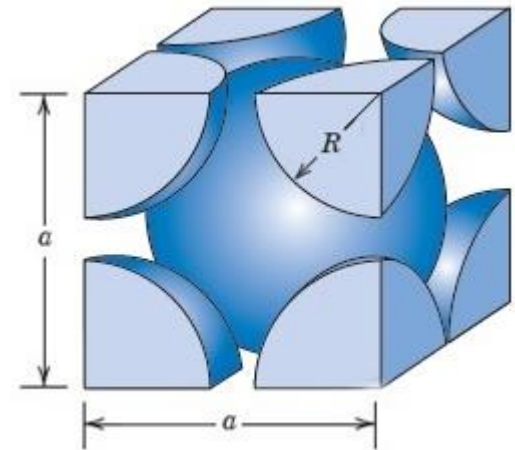
APF of simple cubic:

$$APF = \frac{N_{atoms} \times V_{atom}}{V_{unit\ cell}} = \frac{1 \times \left(\frac{4}{3}\pi R^3\right)}{(2R)^3} = 0.5236$$

APF of BCC:

$$APF = \frac{N_{atoms} \times V_{atom}}{V_{unit\ cell}} = \frac{2 \times \left(\frac{4}{3}\pi R^3\right)}{\left(\frac{4R}{\sqrt{3}}\right)^3} = 0.680174762$$

$$V_{unit\ cell}, a^2 + a^2 + a^2 = (4R)^2, a = \frac{4R}{\sqrt{3}}$$

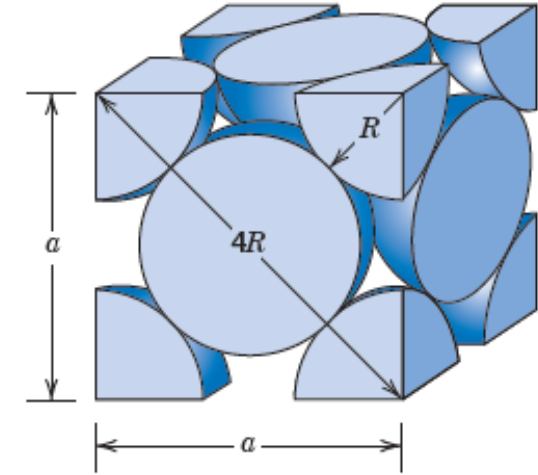


APF of FCC:

$$APF = \frac{N_{atoms} \times V_{atom}}{V_{unit\ cell}} = \frac{4 \times \left(\frac{4}{3}\pi R^3\right)}{(2\sqrt{2}R)^3} = 0.74048048$$

$$V_{unit\ cell}, a^2 + a^2 = (4R)^2, a = 2\sqrt{2}R$$

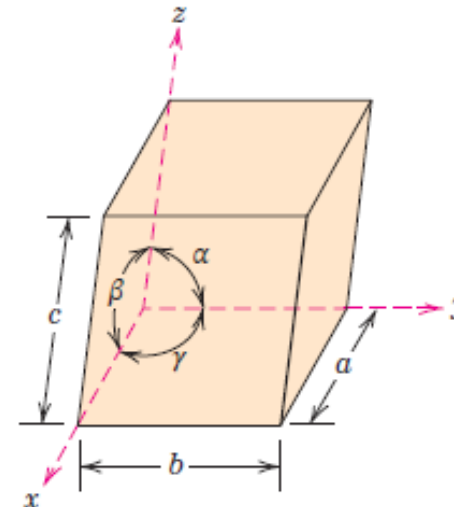
APF of HCP:



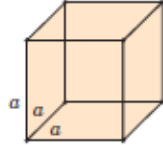
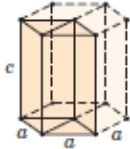
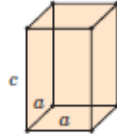
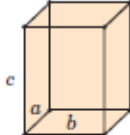
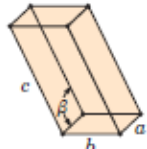
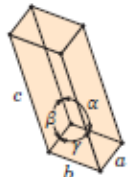
CRYSTAL SYSTEMS

The unit cell geometry is completely defined in terms of six parameters: the three edge lengths a , b , and c and the three interaxial angles α , β , and γ . These are indicated in Figure 6 and are sometimes termed the **lattice parameters** of a crystal structure. On this basis there are seven different possible combinations of a , b , and c and α , β , and γ , each of which represents a distinct **crystal system**. These seven crystal systems are cubic, tetragonal, hexagonal, orthorhombic, rhombohedral (also called trigonal), monoclinic, and triclinic. The lattice parameter relationships and unit cell sketches for each are represented in Table 3.6. The cubic system, for which $a = b = c$ and $\alpha = \beta = \gamma = 90^\circ$, has the greatest degree of symmetry. The least symmetry is displayed by the triclinic system because $a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma$.

Figure 6 A unit cell with x , y , and z coordinate axes, showing axial lengths (a , b , and c) and interaxial angles (α , β , and γ).



23MEE203_Metallurgy and Materials Science

Crystal System	Axial Relationships	Interaxial Angles	Unit Cell Geometry
Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$	
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^\circ, \gamma = 120^\circ$	
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	
Rhombohedral (Trigonal)	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$	
Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	
Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90^\circ \neq \beta$	
Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$	



23MEE203_Metallurgy and Materials Science

Crystallographic Points, Directions, and Planes

Point coordinates:

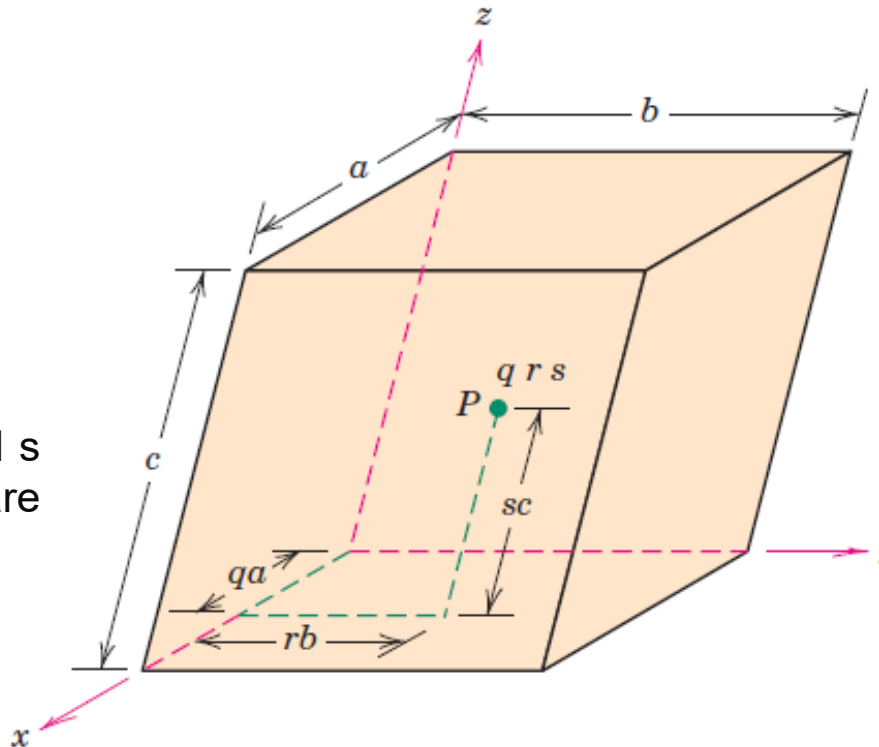
Sometimes it is necessary to specify a lattice position within a unit cell. Lattice position is defined in terms of three *lattice position coordinates*, which are associated with the x , y , and z axes—we have chosen to label these coordinates as P_x , P_y , and P_z . Coordinate specifications are possible using three *point indices*: q , r , and s . These indices are fractional multiples of a , b , and c unit cell lengths—that is, q is some fractional length of a along the x axis, r is some fractional length of b along the y axis, and similarly for s . In other words, lattice position coordinates (i.e., the P s) are equal to the products of their respective point indices and the unit cell edge lengths—viz.

$$P_x = qa$$

$$P_y = rb$$

$$P_z = sc$$

Figure 7 The manner in which the q , r , and s indices at point P within the unit cell are determined.

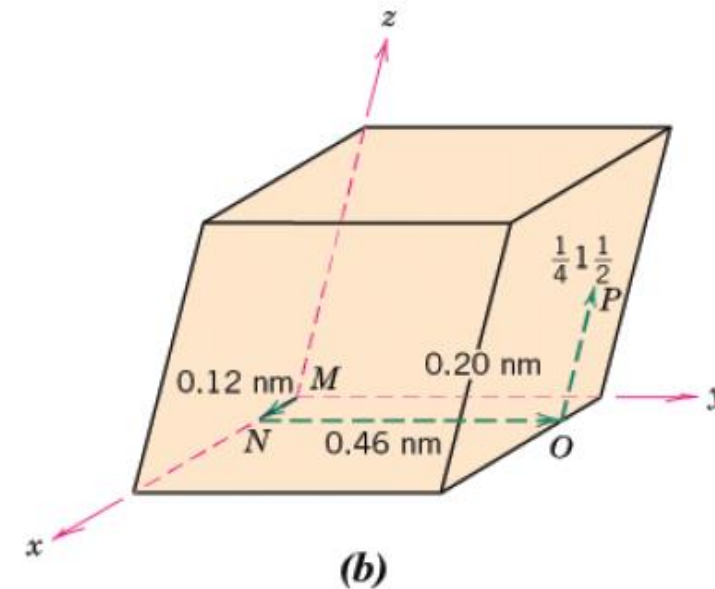
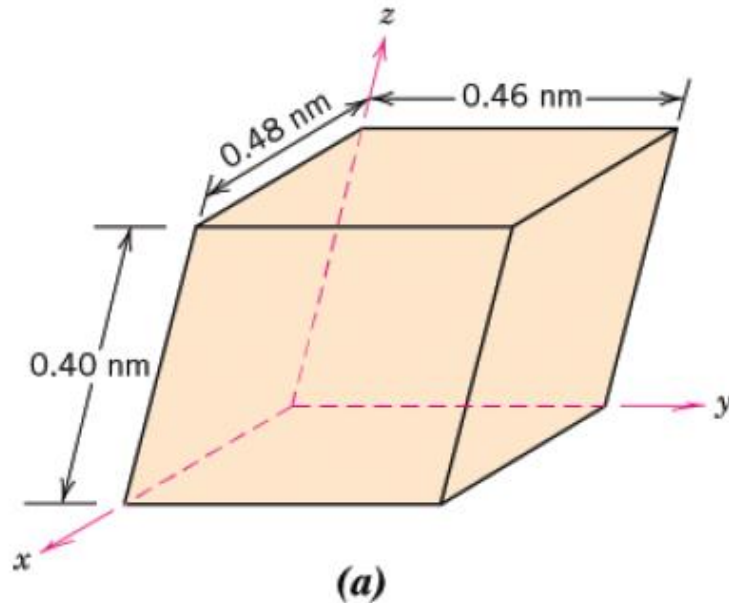


23MEE203_Metallurgy and Materials Science

Example:

Location of Point Having Specified Coordinates

For the unit cell shown in the accompanying sketch (a), locate the point having indices $\frac{1}{4}, 1, \frac{1}{2}$



$$P_x = qa = \frac{1}{4} \times a = \frac{1}{4} \times (0.48 \text{ nm}) = 0.12 \text{ nm}$$

$$P_y = rb = (1) \times b = 1 \times (0.46 \text{ nm}) = 0.46 \text{ nm}$$

$$P_z = sc = \frac{1}{2} \times c = \frac{1}{2} \times (0.40 \text{ nm}) = 0.20 \text{ nm}$$



Crystallographic directions

A *crystallographic direction* is defined as a line directed between two points, or a *vector*.

The following steps are used to determine the three directional indices:

1. A right-handed x-y-z coordinate system is first constructed. As a matter of convenience, its origin may be located at a unit cell corner.
2. The coordinates of two points that lie on the direction vector (referenced to the coordinate system) are determined—for example, for the vector tail, point 1: x_1 , y_1 , and z_1 ; whereas for the vector head, point 2: x_2 , y_2 , and z_2 .
3. Tail point coordinates are subtracted from head point components—that is, $x_2 - x_1$, $y_2 - y_1$, and $z_2 - z_1$.
4. These coordinate differences are then normalized in terms of (i.e., divided by) their respective a , b , and c lattice parameters—that is,

$$\frac{x_2 - x_1}{a}, \frac{y_2 - y_1}{b}, \frac{z_2 - z_1}{c}$$

which yields a set of three numbers.

5. If necessary, these three numbers are multiplied or divided by a common factor to reduce them to the smallest integer values.
6. The three resulting indices, not separated by commas, are enclosed in square brackets, thus: $[uvw]$. The u , v , and w integers correspond to the normalized coordinate differences referenced to the x , y , and z axes, respectively.



23MEE203_Metallurgy and Materials Science

In summary, the u , v , and w indices may be determined using the following equations:

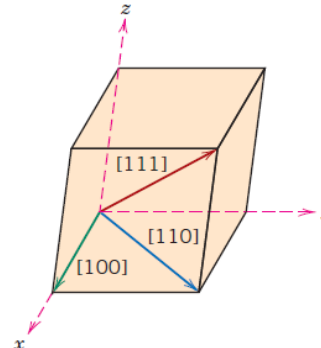
$$u = n \left(\frac{x_2 - x_1}{a} \right)$$

$$v = n \left(\frac{y_2 - y_1}{b} \right)$$

$$w = n \left(\frac{z_2 - z_1}{c} \right)$$

- In these expressions, n is the factor that may be required to reduce u , v , and w to integers.
- For each of the three axes, there are both positive and negative coordinates. Thus, negative indices are also possible, which are represented by a bar over the appropriate index.
- For example, the $[1\bar{1}1]$ direction has a component in the $-y$ direction. Also, changing the signs of all indices produces an antiparallel direction; that is, $[\bar{1}1\bar{1}]$ is directly opposite to $[1\bar{1}1]$.
- The $[100]$, $[110]$, and $[111]$ directions are common ones; they are drawn in the unit cell shown in Figure 8.

Figure 8 The $[100]$, $[110]$, and $[111]$ directions within a unit cell.



EXAMPLE

Construction of a Specified Crystallographic Direction

Within the following unit cell draw a $[1\bar{1}0]$ direction with its tail located at the origin of the coordinate system, point O.

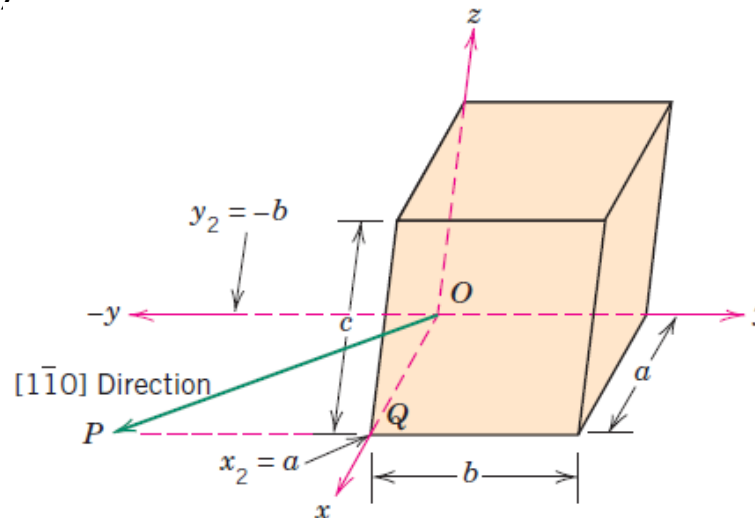
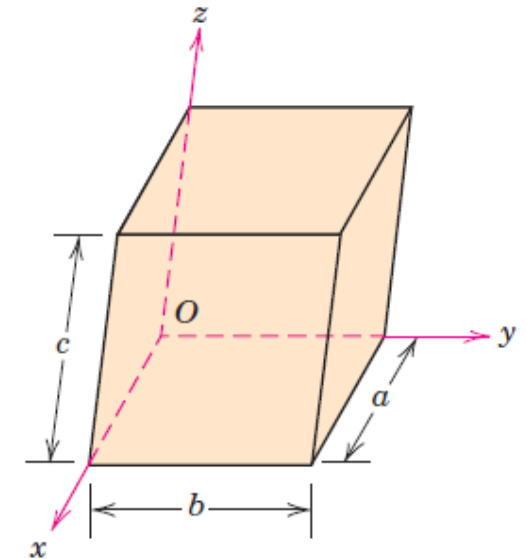
From this $[1\bar{1}0]$ direction we have, $u = 1$, $v = -1$, $w = 0$. As given tail located at the origin thus we have $x_1 = 0a$, $y_1 = 0b$, $z_1 = 0c$.

To determine x_2 , y_2 , z_2 taking n value 1

$$x_2 = ua + x_1 = (1)(a) + 0a = a$$

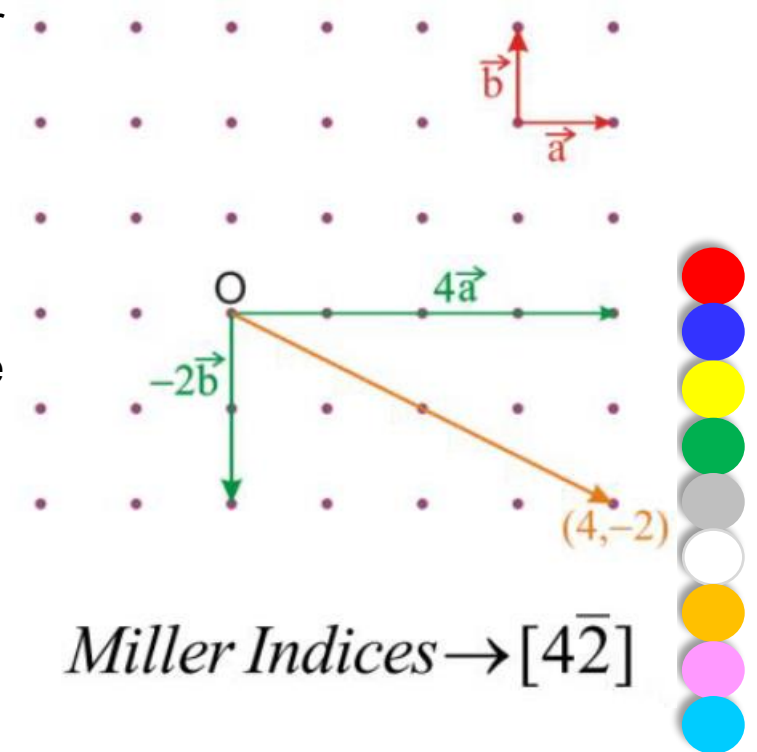
$$y_2 = vb + y_1 = (-1)(b) + 0b = -b$$

$$z_2 = wc + z_1 = (0)(c) + 0c = 0c$$



Miller's indices

- Miller indices are used to specify directions and planes.
- These directions and planes could be in lattices or in crystals.
- The number of indices will match with the dimension of the lattice or the crystal.
- E.g. in 1D there will be 1 index and 2D there will be two indices etc.
- (h,k,l) represents a point..
- Negative numbers/directions are denoted with a bar on top of the number.
- $[hkl]$ represents a direction.
- $\langle hkl \rangle$ represents a family of directions.
- (hkl) represents a plane.
- $\{hkl\}$ represents a family of planes



Miller Indices $\rightarrow [4\bar{2}]$



The orientations of planes for a crystal structure are represented in a similar manner. Again, the unit cell is the basis, with the three-axis coordinate system as represented in Figure 9. In all but the hexagonal crystal system, crystallographic planes are specified by three **Miller indices** as (hkl) . Any two planes parallel to each other are equivalent and have identical indices. The procedure used to determine the h , k , and l index numbers is as follows:

1. If the plane passes through the selected origin, either another parallel plane must be constructed within the unit cell by an appropriate translation, or a new origin must be established at the corner of another unit cell.
2. At this point, the crystallographic plane either intersects or parallels each of the three axes. The coordinate for the intersection of the crystallographic plane with each of the axes is determined (referenced to the origin of the coordinate system). These intercepts for the x , y , and z axes will be designed by A , B , and C , respectively.
3. The reciprocals of these numbers are taken. A plane that parallels an axis is considered to have an infinite intercept and therefore a zero index.
4. The reciprocals of the intercepts are then normalized in terms of (i.e., multiplied by) their respective a , b , and c lattice parameters. That is, $\frac{a}{A} \frac{b}{B} \frac{c}{C}$
5. If necessary, these three numbers are changed to the set of smallest integers by multiplication or by division by a common factor.



6. Finally, the integer indices, not separated by commas, are enclosed within parentheses, thus: (hkl) . The h , k , and l integers correspond to the normalized intercept reciprocals referenced to the x , y , and z axes, respectively.

In summary, the h , k , and l indices may be determined using the following equations:

$$h = \frac{na}{A}$$
$$k = \frac{nb}{B}$$
$$l = \frac{nc}{C}$$

In these expressions, n is the factor that may be required to reduce h , k , and l to integers.

An intercept on the negative side of the origin is indicated by a bar or minus sign positioned over the appropriate index. Furthermore, reversing the directions of all indices specifies another plane parallel to, on the opposite side of, and equidistant from the origin.



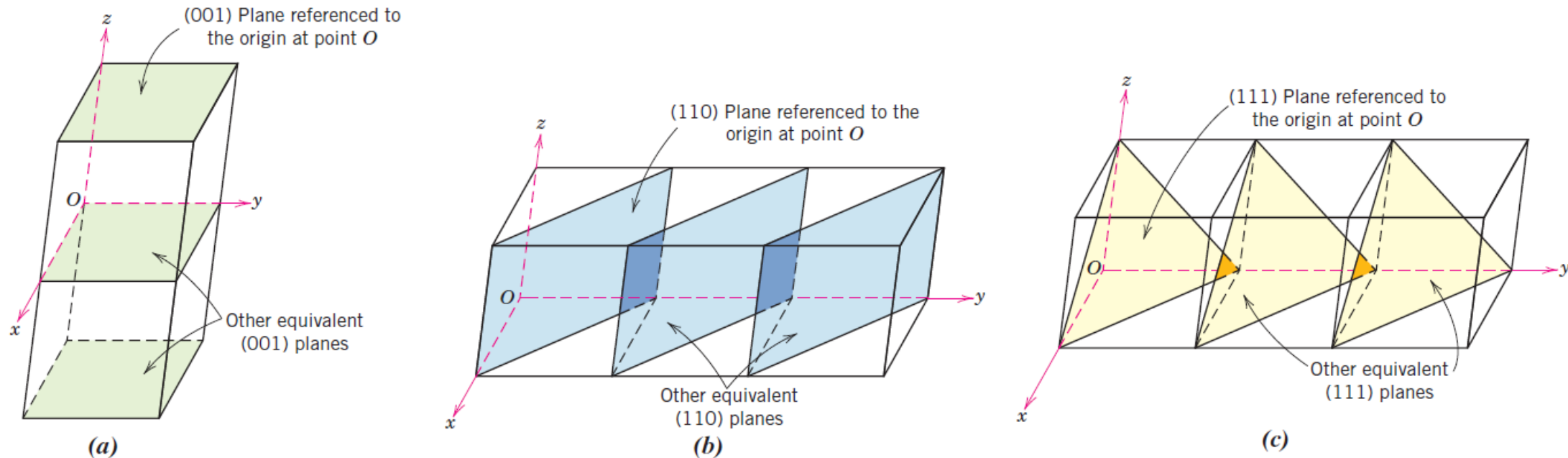


Figure 9 Representations of a series each of the (a) (001) , (b) (110) , and (c) (111) crystallographic planes.



EXAMPLE

Construction of a Specified Crystallographic Plane

Construct a (101) plane within the following unit cell.

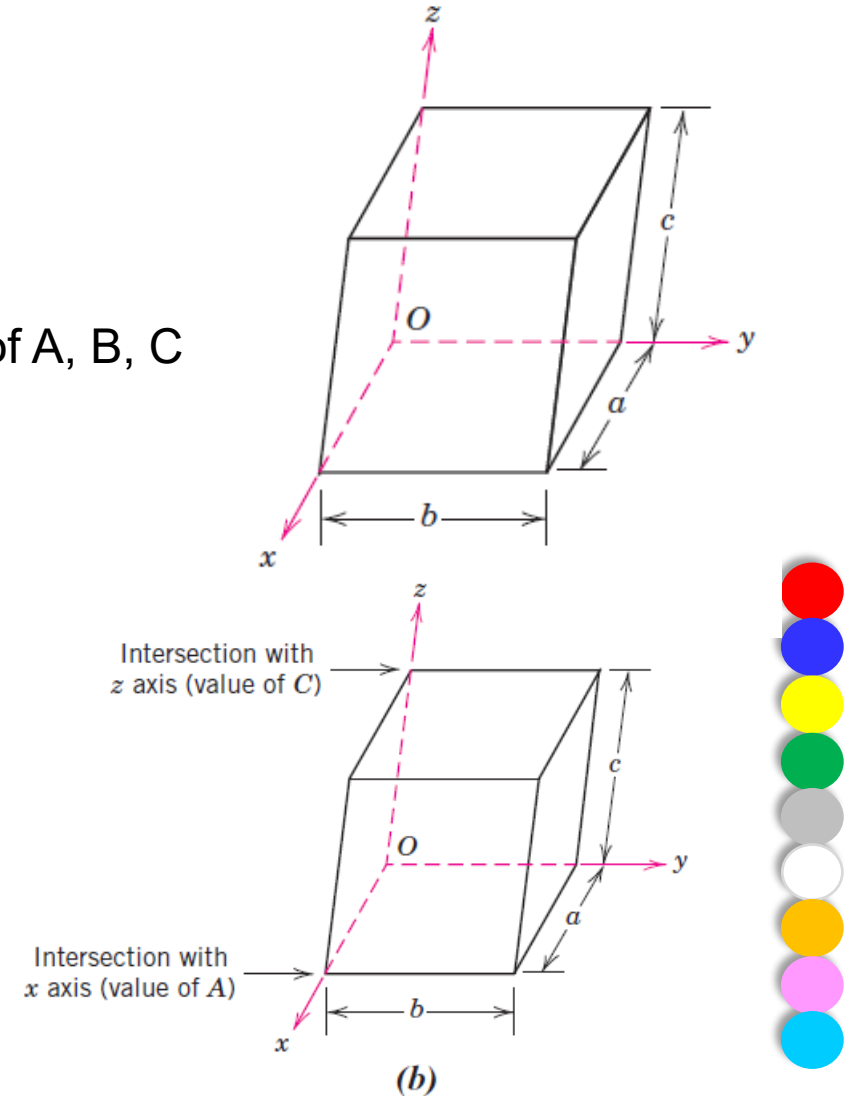
From (101) plane we have, $h = 1$, $k = 0$, $l = 1$. Hence to determination of A, B, C
Take n value 1 then

$$h = \frac{na}{A}, A = \frac{na}{h} = \frac{1a}{1} = a$$

$$k = \frac{nb}{B}, B = \frac{nb}{k} = \frac{1b}{0} = \infty b$$

$$l = \frac{nc}{C}, C = \frac{nc}{l} = \frac{1c}{1} = c$$

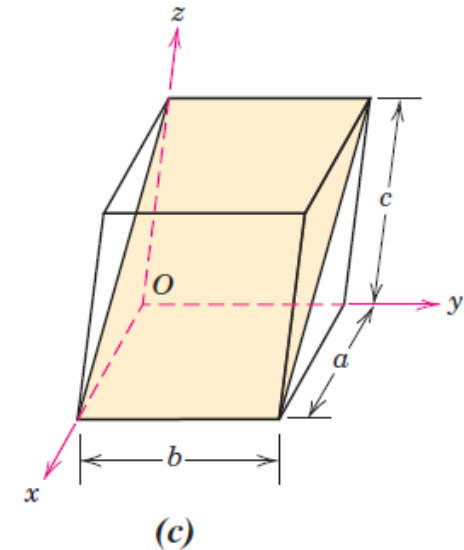
- Thus, this (101) plane intersects the x axis at a (because $A = a$), it parallels the y axis (because $B = \infty b$), and intersects the z axis at c . On the unit cell shown in (b) are noted the locations of the intersections for this plane.



23MEE203_Metallurgy and Materials Science

- The only plane that parallels the y axis and intersects the x and z axes at axial a and c coordinates, respectively, is shown in (c).
- Note that the representation of a crystallographic plane referenced to a unit cell is by lines drawn to indicate intersections of this plane with unit cell faces (or extensions of these faces). The following guides are helpful with representing crystallographic planes:

- If two of the h , k , and l indices are zeros [as with (100)], the plane will parallel one of the unit cell faces (per Figure 9a).
- If one of the indices is a zero [as with (110)], the plane will be a parallelogram, having two sides that coincide with opposing unit cell edges (or edges of adjacent unit cells) (per Figure 9b).
- If none of the indices is zero [as with (111)], all intersections will pass through unit cell faces (per Figure 9c).



Linear and planar densities

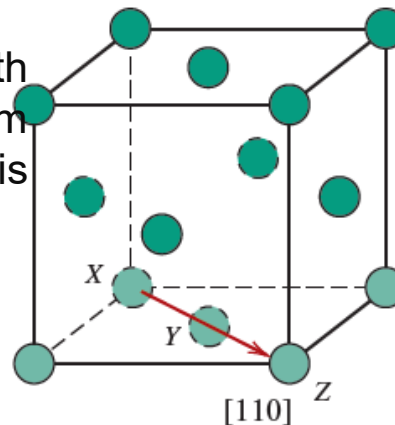
Previously we have discussed the equivalency of nonparallel crystallographic directions and planes. Directional equivalency is related to *linear density* in the sense that, for a particular material, equivalent directions have identical linear densities. The corresponding parameter for crystallographic planes is *planar density*, and planes having the same planar density values are also equivalent.

Linear density (LD) is defined as the number of atoms per unit length whose centers lie on the direction vector for a specific crystallographic direction; that is,

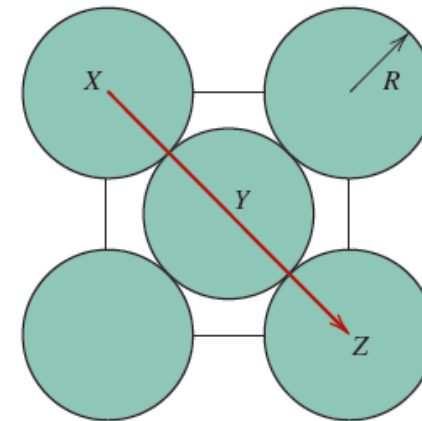
$$LD = \frac{\text{number of atoms centered on direction vector}}{\text{length of direction vector}}$$

The units of linear density are reciprocal length (e.g., nm^{-1} , m^{-1}).

Figure 10 (a) Reduced-sphere FCC unit cell with the $[110]$ direction indicated. (b) The bottom face-plane of the FCC unit cell in (a) on which is shown the atomic spacing in the $[110]$ direction, through atoms labeled X, Y, and Z.



(a)



(b)



19MEE204_Metallurgy and Material Science

- For example, let us determine the linear density of the [110] direction for the FCC crystal structure.
- An FCC unit cell (reduced sphere) and the [110] direction therein are shown in Figure 10a. Represented in Figure 10b are five atoms that lie on the bottom face of this unit cell; here the [110] direction vector passes from the center of atom X, through atom Y, and finally to the center of atom Z. With regard to the numbers of atoms, it is necessary to take into account the sharing of atoms with adjacent unit.
- Each of the X and Z corner atoms is also shared with one other adjacent unit cell along this [110] direction (i.e., one-half of each of these atoms belongs to the unit cell being considered), whereas atom Y lies entirely within the unit cell.
- Thus, there is an equivalence of two atoms along the [110] direction vector in the unit cell.
- Now, the direction vector length is equal to $4R$ (Figure 10b); thus, the [110] linear density for FCC is

$$LD_{110} = \frac{2 \text{ atoms}}{4R} = \frac{1}{2R}$$



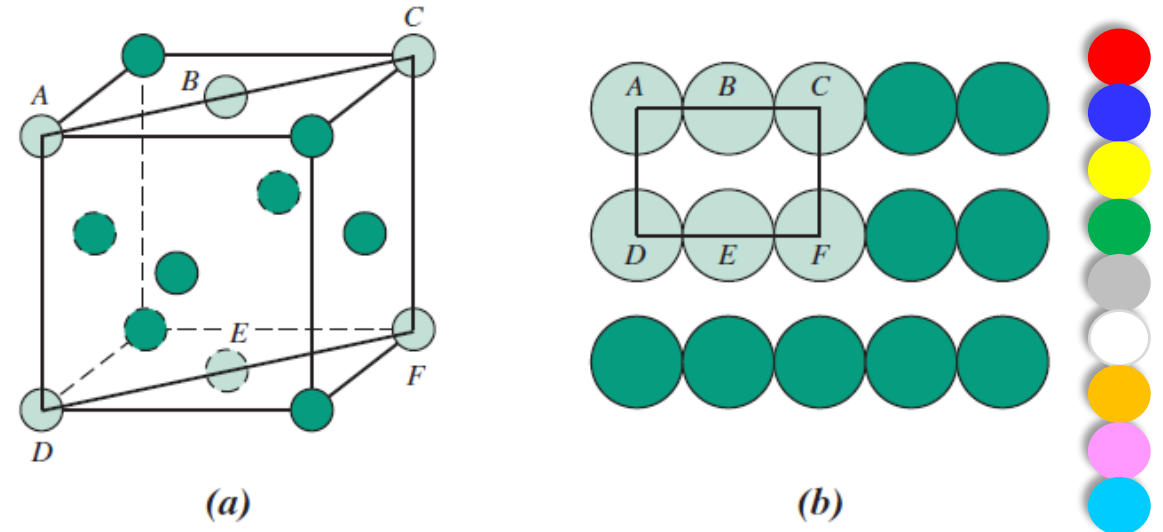
Planer density:

In an analogous manner, *planar density* (PD) is taken as the number of atoms per unit area that are centered on a particular crystallographic plane, or

$$PD = \frac{\text{number of atoms centered on a plane}}{\text{area of plane}}$$

The units for planar density are reciprocal area (e.g., nm⁻², m⁻²).

Figure 11 (a) Reduced-sphere FCC unit cell with the (110) plane. (b) Atomic packing of an FCC (110) plane. Corresponding atom positions from (a) are indicated.



23MEE203_Metallurgy and Materials Science

For example, consider the section of a (110) plane within an FCC unit cell as represented in Figures 10a and 10b. Although six atoms have centers that lie on this plane (Figure 10b), only one-quarter of each of atoms *A*, *C*, *D*, and *F* and one-half of atoms *B* and *E*, for a total equivalence of just 2 atoms, are on that plane.

Furthermore, the area of this rectangular section is equal to the product of its length and width. From Figure 10b, the length (horizontal dimension) is equal to $4R$, whereas the width (vertical dimension) is equal to $2R\sqrt{2}$ because it corresponds to the FCC unit cell edge length. Thus, the area of this planar region is $(4R)(2R\sqrt{2}) = 8R^2\sqrt{2}$, and the planar density is determined as follows:

$$PD_{110} = \frac{2 \text{ atoms}}{8R^2\sqrt{2}} = \frac{1}{4R^2\sqrt{2}}$$

Linear and planar densities are important considerations relative to the process of slip—that is, the mechanism by which metals plastically deform. Slip occurs on the most densely packed crystallographic planes and, in those planes, along directions having the greatest atomic packing.



23MEE203_Metallurgy and Materials Science

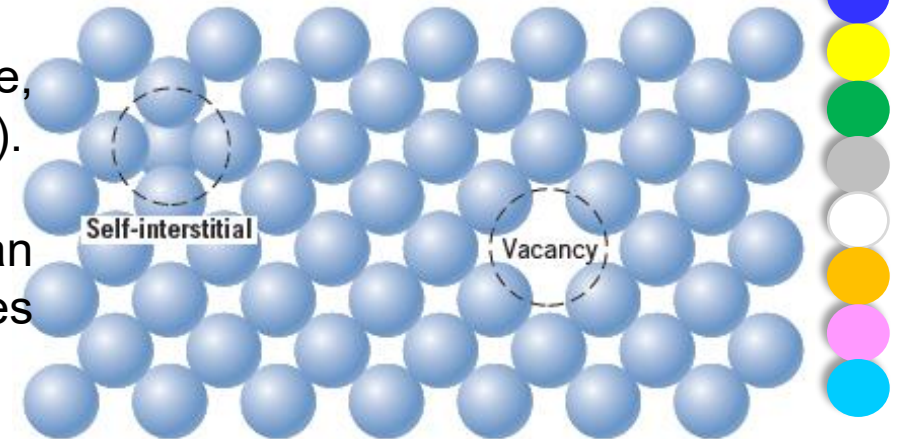
It has been tacitly assumed that perfect order exists throughout crystalline materials on an atomic scale. However, such an idealized solid does not exist; all contain large numbers of various defects or **imperfections**.

A *crystalline defect* refers to a lattice irregularity having one or more of its dimensions on the order of an atomic diameter. Classification of crystalline imperfections is frequently made according to the geometry or dimensionality of the defect.

Several different imperfections need to be discussed, including **point defects** (those associated with one or two atomic positions); linear (or one-dimensional) defects; and interfacial defects, or boundaries, which are two-dimensional.

The simplest of the point defects is a **vacancy**, or vacant lattice site, one normally occupied but from which an atom is missing (Figure 12).

A **self-interstitial** is an atom from the crystal that is crowded into an interstitial site—a small void space that under ordinary circumstances is not occupied.



 **Figure 12** Two-dimensional representations of a vacancy and a self-interstitial.

23MEE203_Metallurgy and Materials Science

Impurity point defects are found in solid solutions, of which there are two types: **substitutional** and **interstitial**. For the substitutional type, solute or impurity atoms replace or substitute for the host atoms (Figure 13).

For interstitial solid solutions, impurity atoms fill the voids or interstices among the host atoms (see Figure 13).

A **dislocation** is a **linear** or one-dimensional defect around which some of the atoms are misaligned. One type of dislocation is represented in Figure 14: an extra portion of a plane of atoms, or half-plane, the edge of which terminates within the crystal.

This is termed an **edge dislocation**; it is a linear defect that centers on the line that is defined along the end of the extra half-plane of atoms. This is sometimes termed the **dislocation line**, which, for the edge dislocation in Figure 14, is perpendicular to the plane of the page.

Sometimes the edge dislocation in Figure 14 is represented by the symbol $\perp$, which also indicates the position of the dislocation line. An edge dislocation may also be formed by an extra half-plane of atoms that is included in the bottom portion of the crystal; its designation is a T.

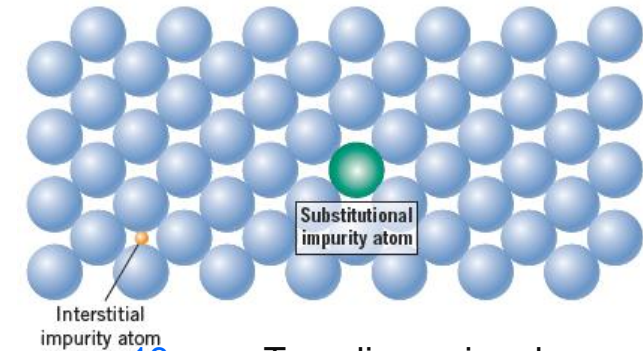


Figure 13
Two-dimensional schematic representations of substitutional and interstitial impurity atoms

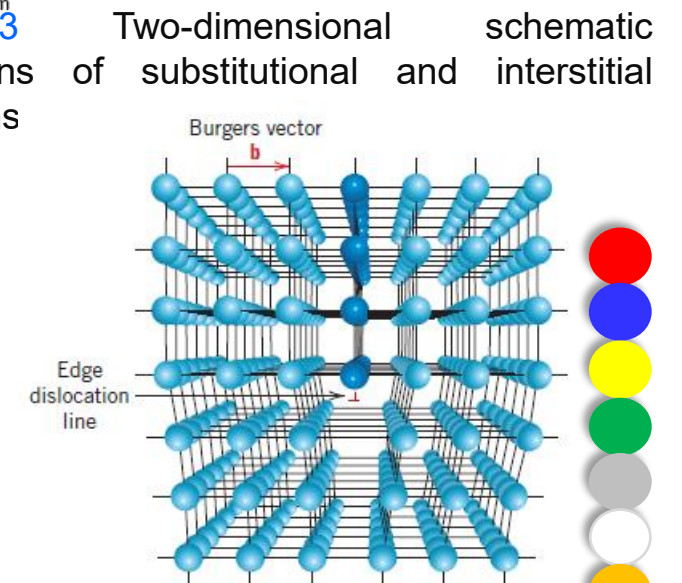


Figure 14 The atom positions around an edge dislocation; extra half-plane of atoms shown in perspective.

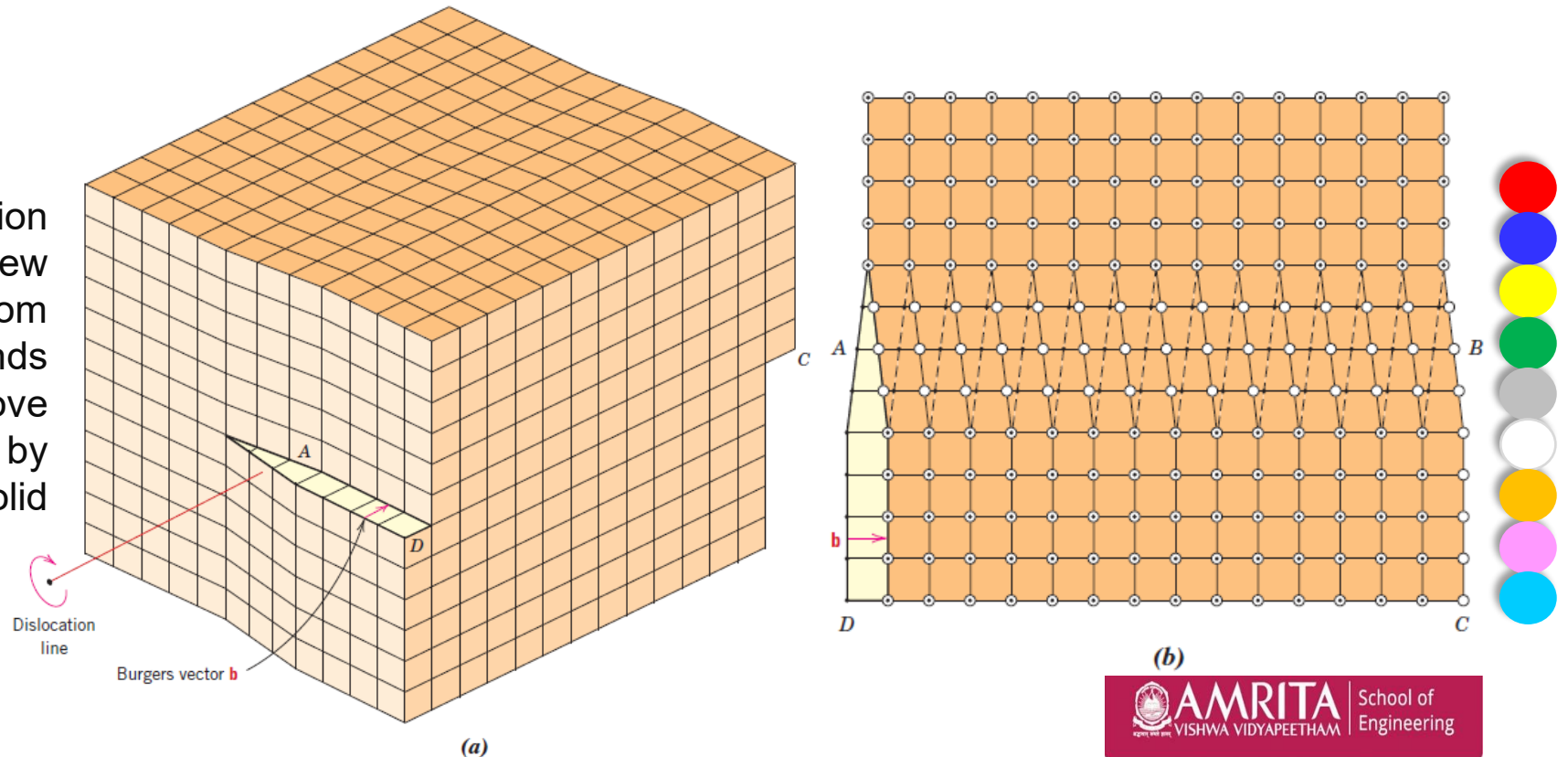


23MEE203_Metallurgy and Materials Science

Another type of dislocation, called a **screw dislocation**, may be thought of as being formed by a shear stress that is applied to produce the distortion shown in Figure 15a:

The upper front region of the crystal is shifted one atomic distance to the right relative to the bottom portion. The atomic distortion associated with a screw dislocation is also linear and along a dislocation line, line *AB* in Figure 15b. The screw dislocation derives its name from the spiral or helical path or ramp that is traced around the dislocation line by the atomic planes of atoms. Sometimes the symbol is used to designate a screw dislocation.

Figure 15 (a) A screw dislocation within a crystal. (b) The screw dislocation in (a) as viewed from above. The dislocation line extends along line *AB*. Atom positions above the slip plane are designated by open circles, those below by solid circles.



Interfacial defects

Interfacial defects are boundaries that have two dimensions and normally separate regions of the materials that have different crystal structures and/or crystallographic orientations. These imperfections include external surfaces, grain boundaries, phase boundaries, twin boundaries, and stacking faults.

External Surfaces:

One of the most obvious boundaries is the external surface, along which the crystal structure terminates. Surface atoms are not bonded to the maximum number of nearest neighbors and are therefore in a higher-energy state than the atoms at interior positions.

The bonds of these surface atoms that are not satisfied give rise to a surface energy, expressed in units of energy per unit area (J/m^2 or erg/cm^2).

To reduce this energy, materials tend to minimize, if at all possible, the total surface area.

For example, liquids assume a shape having a minimum area—the droplets become spherical. Of course, this is not possible with solids, which are mechanically rigid.



Grain Boundaries

- A grain boundary is represented schematically from an atomic perspective in Figure 16.
- Within the boundary region, which is probably just several atom distances wide, there is some atomic mismatch in a transition from the crystalline orientation of one grain to that of an adjacent one.
- Various degrees of crystallographic misalignment between adjacent grains are possible (Figure 16).
- When this orientation mismatch is slight, on the order of a few degrees, then the term *small- (or low-) angle grain boundary* is used.

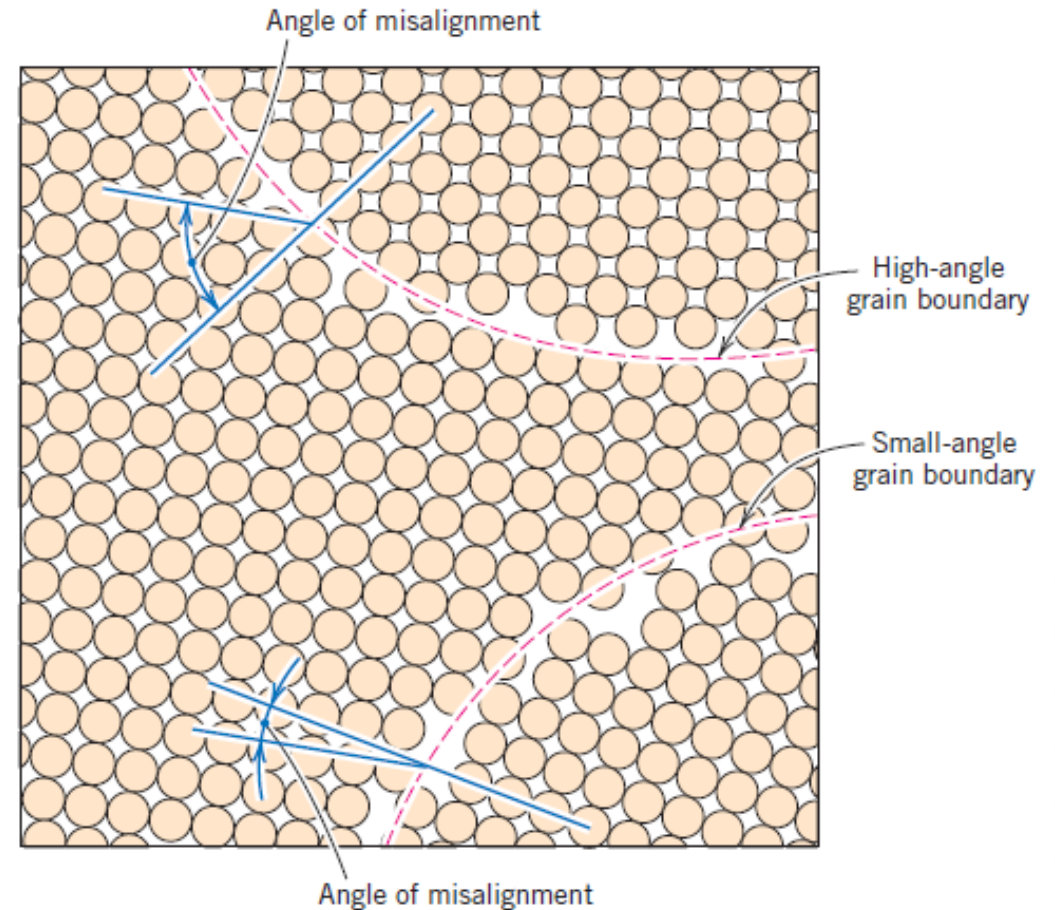
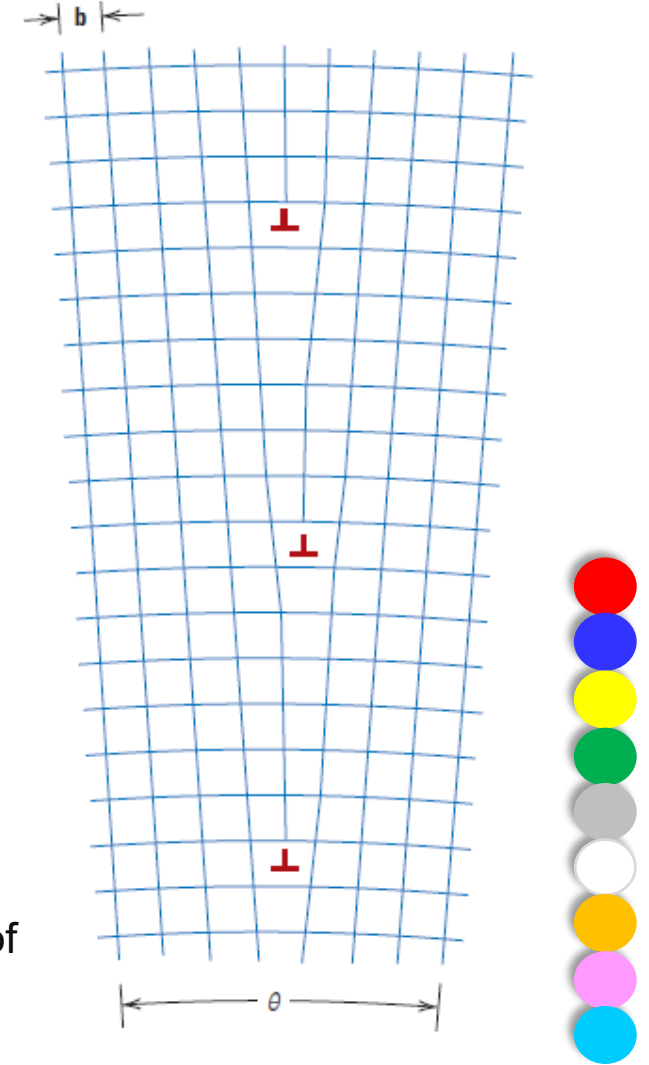


Figure 16 Schematic diagram showing small- and high-angle grain boundaries and the adjacent atom positions.



- These boundaries can be described in terms of dislocation arrays.
- One simple small-angle grain boundary is formed when edge dislocations are aligned in the manner of Figure 17.
- This type is called a *tilt boundary*; the angle of misorientation, θ , is also indicated in the figure.
- When the angle of misorientation is parallel to the boundary, a *twist boundary* results, which can be described by an array of screw dislocations.

Figure 17 Demonstration of how a tilt boundary having an angle of misorientation θ results from an alignment of edge dislocations.



Phase Boundaries

Phase boundaries exist in multiphase materials, in which a different phase exists on each side of the boundary; additionally, each of the constituent phases has its own distinctive physical and/or chemical characteristics. As we shall see in subsequent chapters, phase boundaries play an important role in determining the mechanical characteristics of some multiphase metal alloys.

Twin Boundaries

A *twin boundary* is a special type of grain boundary across which there is a specific mirror lattice symmetry; that is, atoms on one side of the boundary are located in mirror-image positions to those of the atoms on the other side (Figure 18). The region of material between these boundaries is appropriately termed a *twin*. Twins result from atomic displacements that are produced from applied mechanical shear forces (**mechanical twins**) and also during annealing heat treatments following deformation (**annealing twins**). Twinning occurs on a definite crystallographic plane and in a specific direction, both of which depend on the crystal structure. **Annealing twins are typically found in metals that have the FCC crystal structure**, whereas **mechanical twins are observed in BCC and HCP metals**. The role of mechanical twins in the deformation process.

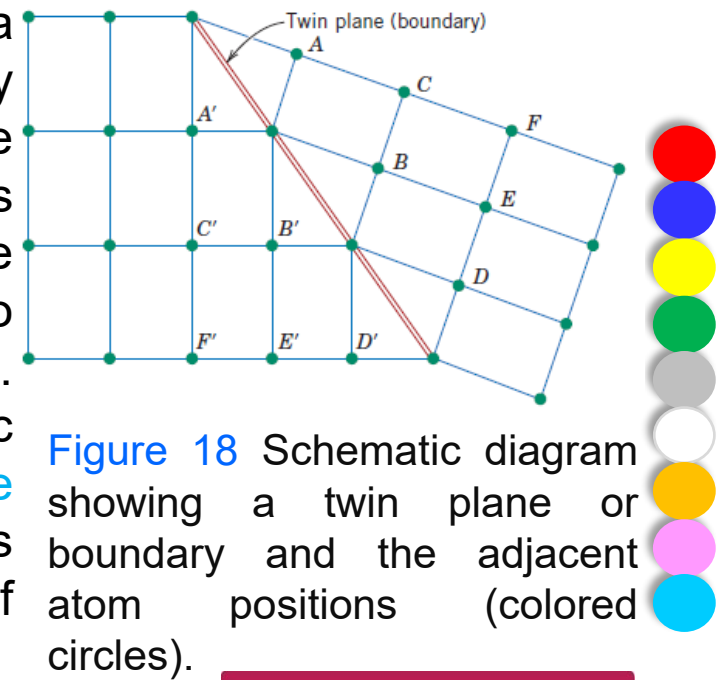


Figure 18 Schematic diagram showing a twin plane or boundary and the adjacent atom positions (colored circles).



Basic concepts of dislocations

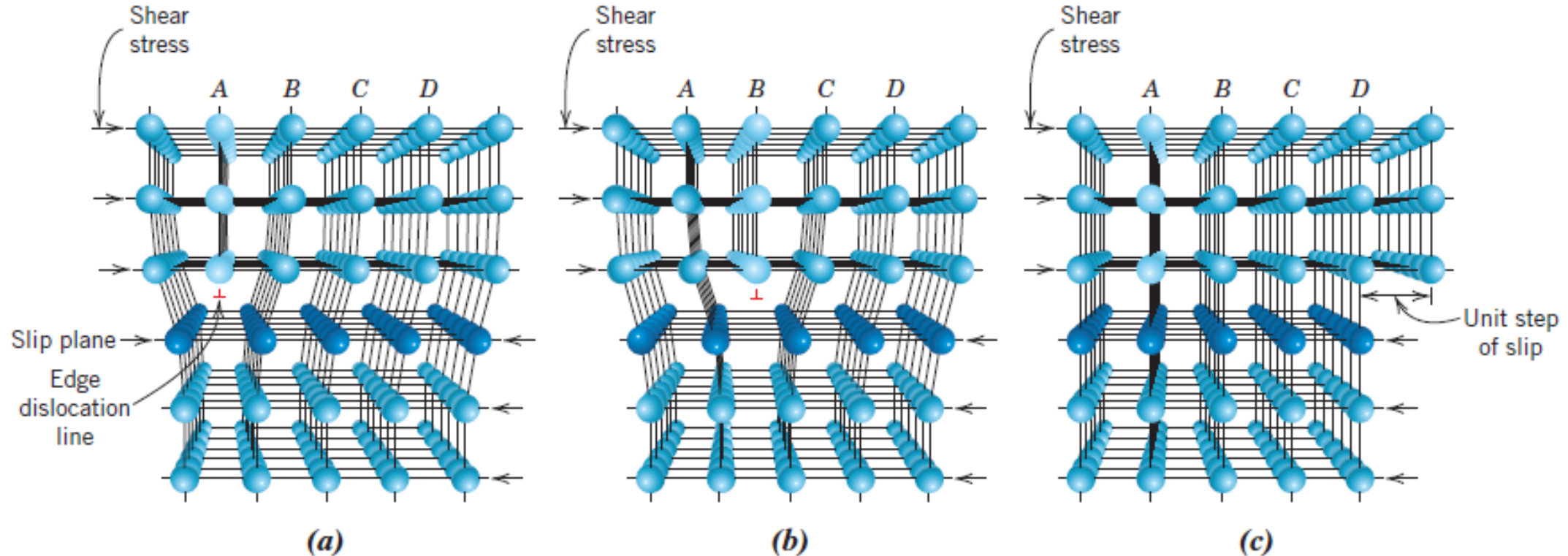


Figure 24 Atomic rearrangements that accompany the motion of an edge dislocation as it moves in response to an applied shear stress. (a) The extra half-plane of atoms is labeled A. (b) The dislocation moves one atomic distance to the right as A links up to the lower portion of plane B; in the process, the upper portion of B becomes the extra half-plane. (c) A step forms on the surface of the crystal as the extra half-plane exits.



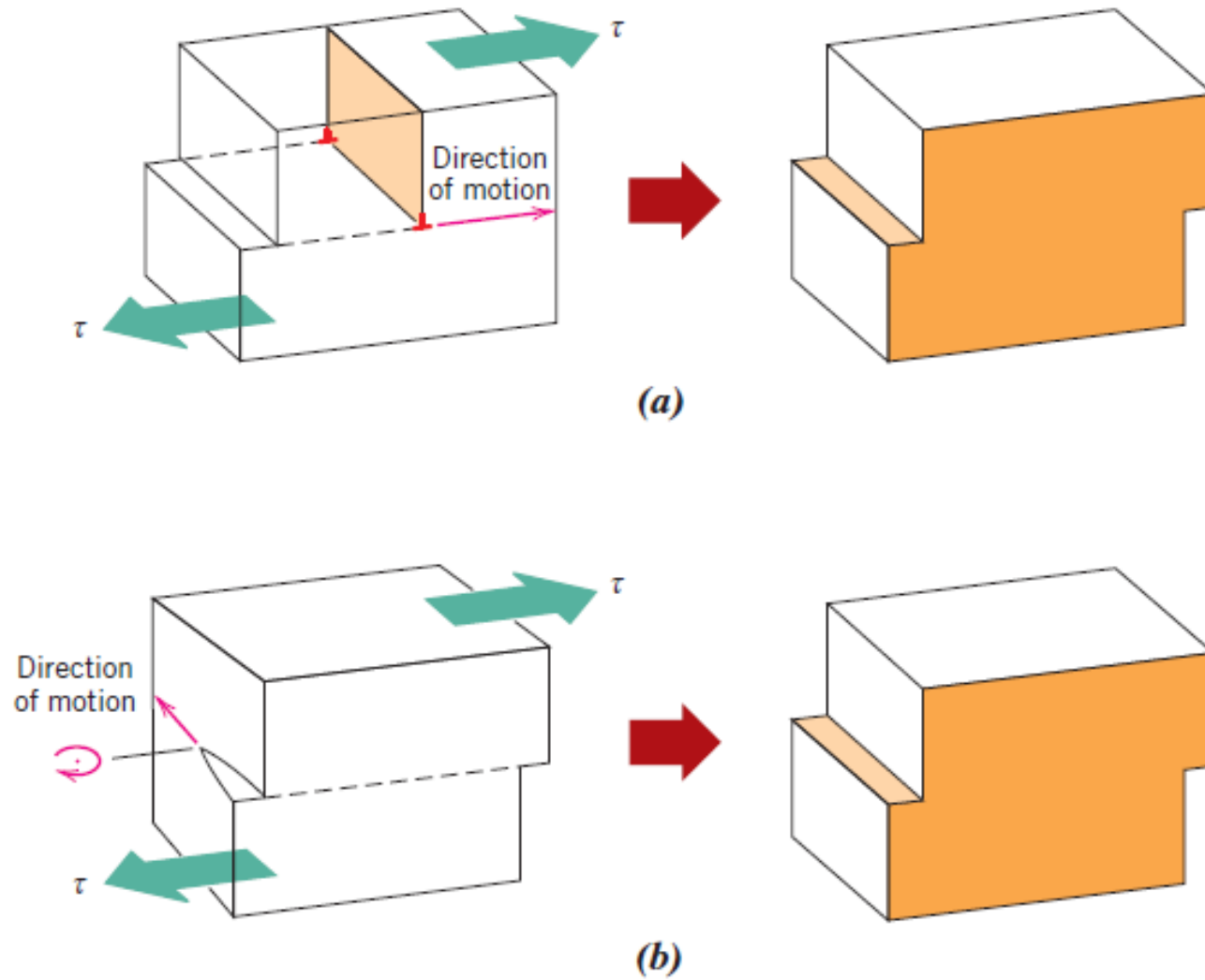


Figure 25 The formation of a step on the surface of a crystal by the motion of (a) an edge dislocation and (b) a screw dislocation.



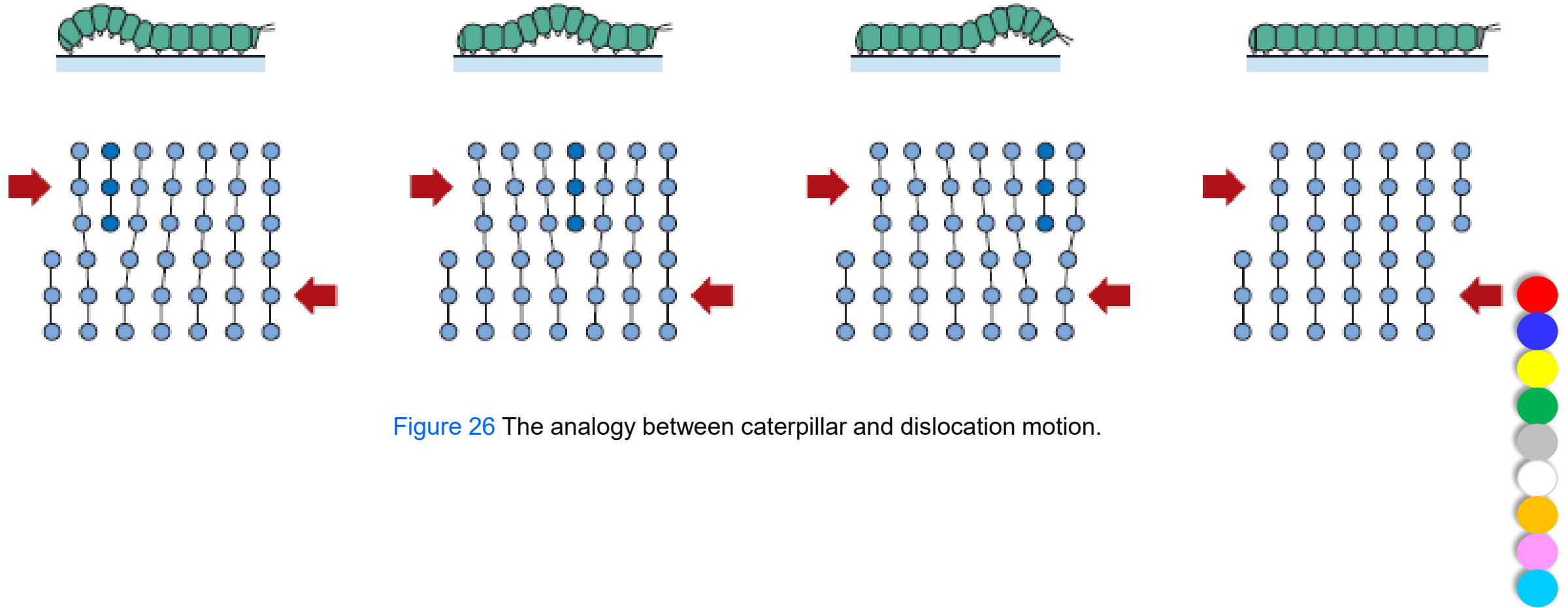


Figure 26 The analogy between caterpillar and dislocation motion.



Dislocations and plastic deformation

Characteristics of dislocations

Several characteristics of dislocations are important with regard to the mechanical properties of metals. These include strain fields that exist around dislocations, which are influential in determining the mobility of the dislocations, as well as their ability to multiply.

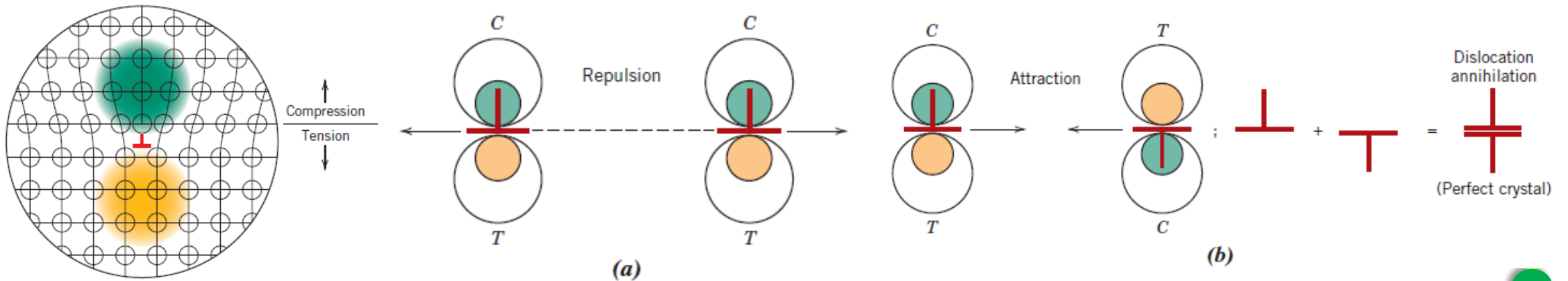


Figure 27 Regions of compression (green) and tension (yellow) located around an edge dislocation.

Figure 28 (a) Two edge dislocations of the same sign and lying on the same slip plane exert a repulsive force on each other; C and T denote compression and tensile regions, respectively. (b) Edge dislocations of opposite sign and lying on the same slip plane exert an attractive force on each other. Upon meeting, they annihilate each other and leave a region of perfect crystal.



Slip phenomenon

Slip systems

- Dislocations do not move with the same degree of ease on all crystallographic planes of atoms and in all crystallographic directions.
- Typically, there is a preferred plane, and in that plane there are specific directions along which dislocation motion occurs. This plane is called the *slip plane*; it follows that the direction of movement is called the *slip direction*.
- This combination of the slip plane and the slip direction is termed the **slip system**. The slip system depends on the crystal structure of the metal and is such that the atomic distortion that accompanies the motion of a dislocation is a minimum.
- For a particular crystal structure, the slip plane is the plane that has the densest atomic packing—that is, has the greatest planar density. The slip direction corresponds to the direction in this plane that is most closely packed with atoms—that is, has the highest linear density.



Slip in single crystals

Slip in single crystals

Even though an applied stress may be pure tensile (or compressive), shear components exist at all but parallel or perpendicular alignments to the stress direction.

These are termed **resolved shear stresses**, and their magnitudes depend not only on the applied stress, but also on the orientation of both the slip plane and direction within that plane.

Let ϕ represent the angle between the normal to the slip plane and the applied stress direction, and let λ be the angle between the slip and stress directions, as indicated in Figure 29; it can then be shown that for the resolved shear stress τ_R

$$\tau_R = \sigma \cos \phi \cos \lambda$$

where σ is the applied stress. In general, $\phi + \lambda \neq 90^\circ$ because it need not be the case that the tensile axis, the slip plane normal, and the slip direction all lie in the same plane.

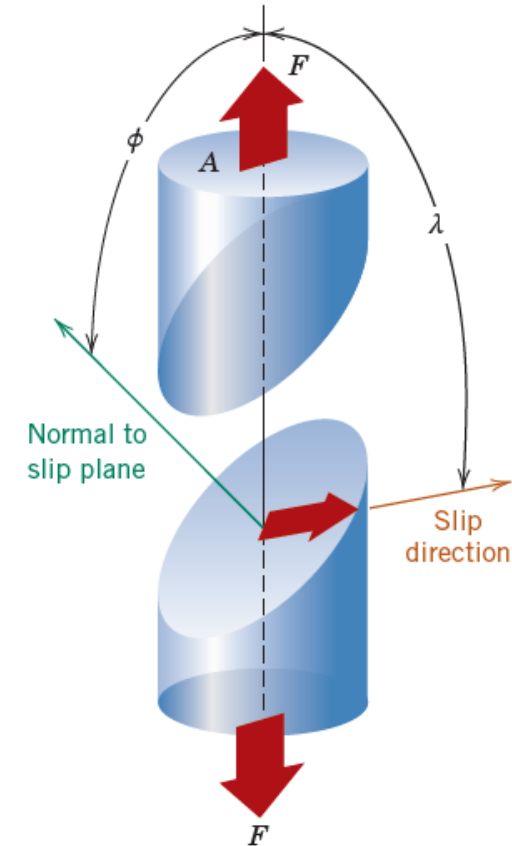


Figure 29 Geometric relationships between the tensile axis, slip plane, and slip direction used in calculating the resolved shear stress for a single crystal.



Slip in single crystals

In response to an applied tensile or compressive stress, slip in a single crystal commences on the most favorably oriented slip system when the resolved shear stress reaches some critical value, termed the **critical resolved shear stress** τ_{crss} ; it represents the minimum shear stress required to initiate slip and is a property of the material that determines when yielding occurs.

The single crystal plastically deforms or yields when $\tau_R(\max) = \tau_{crss}$, and the magnitude of the applied stress required to initiate yielding (i.e., the yield strength σ_y) is

$$\sigma_y = \frac{\tau_{crss}}{(\cos\phi\cos\lambda)_{max}}$$

The minimum stress necessary to introduce yielding occurs when a single crystal is oriented such that $\phi = \lambda = 45^\circ$; under these conditions,

$$\sigma_y = 2\tau_{crss}$$



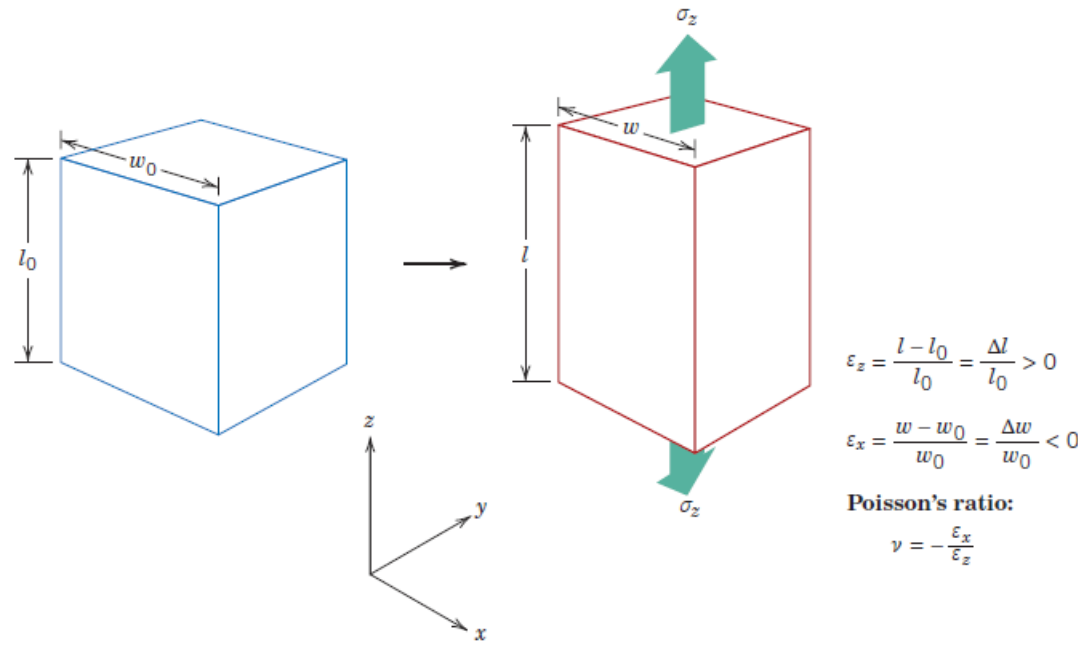
Elastic properties of materials

When a tensile stress is imposed on a metal specimen, an elastic elongation and accompanying strain ϵ_z result in the direction of the applied stress (arbitrarily taken to be the z direction), as indicated in Figure 19. As a result of this elongation, there will be constrictions in the lateral (x and y) directions perpendicular to the applied stress; from these contractions, the compressive strains ϵ_x and ϵ_y may be determined. If the applied stress is uniaxial (only in the z direction) and the material is isotropic, then $\epsilon_x = \epsilon_y$. A parameter termed **Poisson's ratio** ν is defined as the ratio of the lateral and axial strains, or

$$\nu = -\frac{\epsilon_x}{\epsilon_z} = -\frac{\epsilon_y}{\epsilon_z}$$

For virtually all structural materials, ϵ_x and ϵ_z will be of opposite sign; therefore, the negative sign is included in the preceding expression to ensure that ν is positive.

Figure 19 Schematic illustration showing axial (z) elongation (positive strain, ϵ_z) and the lateral (x) contraction (negative strain, ϵ_x) that result from the application of an axial tensile stress (σ_z).



Anelastic properties of materials

- It has been assumed that elastic deformation is time independent— that is, an applied stress produces an instantaneous elastic strain that remains constant over the period of time the stress is maintained.
- It has also been assumed that upon release of the load, the strain is totally recovered—that is, that the strain immediately returns to zero.
- In most engineering materials, however, there will also exist a time-dependent elastic strain component—that is, elastic deformation will continue after the stress application, and upon load release, some finite time is required for complete recovery.
- This time-dependent elastic behavior is known as **anelasticity**, and it is due to time-dependent microscopic and atomistic processes that are attendant to the deformation.



Mechanical Behavior

For most metallic materials, elastic deformation persists only to strains of about 0.002. As the material is deformed beyond this point, the stress is no longer proportional to strain (Hooke's law), and permanent, nonrecoverable, **plastic deformation** or **plastic deformation** occurs. Figure 20a plots schematically the tensile stress–strain behavior in the plastic region for a typical metal. The transition from elastic to plastic is a gradual one for most metals; some curvature results at the onset of plastic deformation, which increases more rapidly with rising stress.

From an atomic perspective, plastic deformation corresponds to the breaking of bonds with original atom neighbors and then the re-forming of bonds with new neighbors as large numbers of atoms or molecules move relative to one another; upon removal of the stress, they do not return to their original positions. This permanent deformation for metals is accomplished by means of a process called *slip*, which involves the motion of dislocations.

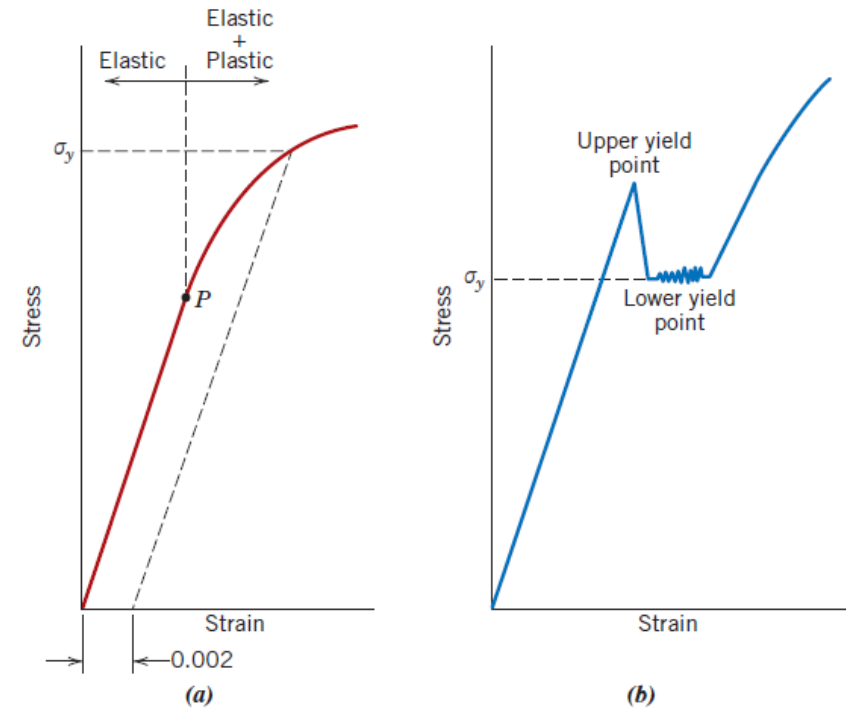


Figure 20 (a) Typical stress–strain behavior for a metal, showing elastic and plastic deformations, the proportional limit P, and the yield strength σ_y , as determined using the 0.002 strain offset method. (b) Representative stress–strain behavior found for some steels, demonstrating the yield point phenomenon.



23MEE203_Metallurgy and Materials Science



Ductility

Ductility is another important mechanical property. It is a measure of the degree of plastic deformation that has been sustained at fracture. A metal that experiences very little or no plastic deformation upon fracture is termed *brittle*. The tensile stress– strain behaviors for both ductile and brittle metals are schematically illustrated in Figure 22.

Ductility may be expressed quantitatively as either *percent elongation* or *percent reduction in area*. Percent elongation (%EL) is the percentage of plastic strain at fracture, or

$$\%EL = \left(\frac{l_f - l_0}{l_0} \right) \times 100$$

where l_f is the fracture length and l_0 is the original gauge length as given earlier.

Percent reduction in area (%RA) is defined as

$$\%RA = \left(\frac{A_f - A_0}{A_0} \right) \times 100$$

where A_0 is the original cross-sectional area and A_f is the cross-sectional area at the point of fracture.

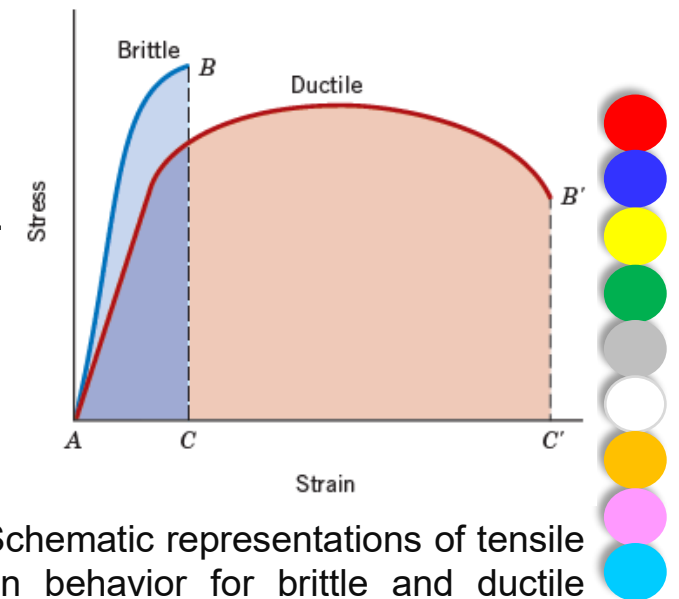


Figure 22 Schematic representations of tensile stress–strain behavior for brittle and ductile metals loaded to fracture.



Resilience

Resilience is the capacity of a material to absorb energy when it is deformed elastically and then, upon unloading, to have this energy recovered. The associated property is the *modulus of resilience*, U_r , which is the strain energy per unit volume required to stress a material from an unloaded state up to the point of yielding.

Computationally, the modulus of resilience for a specimen subjected to a uniaxial tension test is just the area under the engineering stress–strain curve taken to yielding (Figure 23), or

$$U_r = \int_0^{\epsilon_y} \sigma d\epsilon$$

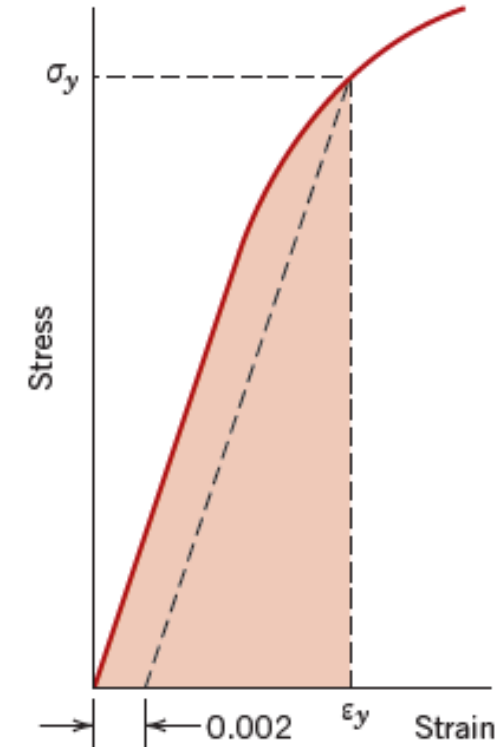


Figure 23 Schematic representation showing how modulus of resilience (corresponding to the shaded area) is determined from the tensile stress–strain behavior of a material.



Toughness

Toughness is a mechanical term that may be used in several contexts. For one, toughness (or more specifically, *fracture toughness*) is a property that is indicative of a material's resistance to fracture when a crack (or other stress-concentrating defect) is present. Because it is nearly impossible (as well as costly) to manufacture materials with zero defects (or to prevent damage during service), fracture toughness is a major consideration for all structural materials.

Another way of defining toughness is as the ability of a material to absorb energy and plastically deform before fracturing. For dynamic (high strain rate) loading conditions and when a notch (or point of stress concentration) is present, *notch toughness* is assessed by using an impact test.

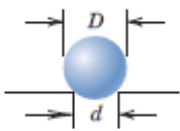
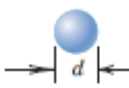
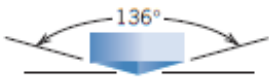
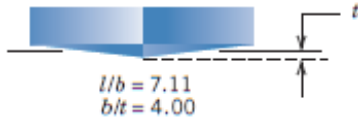
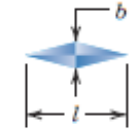
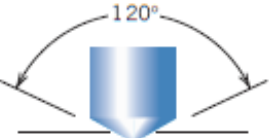
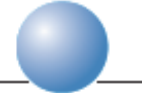
For the static (low strain rate) situation, a measure of toughness in metals (derived from plastic deformation) may be ascertained from the results of a tensile stress–strain test. It is the area under the σ – ϵ curve up to the point of fracture. The units are the same as for resilience (i.e., energy per unit volume of material). For a metal to be tough, it must display both strength and ductility. This is demonstrated in Figure 22, in which the stress–strain curves are plotted for both metal types. Hence, even though the brittle metal has higher yield and tensile strengths, it has a lower toughness than the ductile one, as can be seen by comparing the areas ABC and $AB'C'$ in Figure 22.



23MEE203_Metallurgy and Materials Science

Hardness

Another mechanical property that may be important to consider is **hardness**, which is a measure of a material's resistance to localized plastic deformation (e.g., a small dent or a scratch).

Load	Shape of Indentation				Load	Formula for Hardness Number ^a
	Test	Indenter	Side View	Top View		
500 -3000 kg in 500 kg increment	Brinell	10-mm sphere of steel or tungsten carbide			P	$HB = \frac{2P}{\pi D [D - \sqrt{D^2 - d^2}]}$
	Vickers microhardness	Diamond pyramid			P	$HV = 1.854P/d_1^2$
1-1000g	Knoop microhardness	Diamond pyramid			P	$HK = 14.2P/l^2$
	Rockwell and superficial Rockwell	Diamond cone; $\frac{1}{16}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}$ in. diameter steel or tungsten carbide spheres	 	 	$\left. \begin{array}{l} 60 \text{ kg} \\ 100 \text{ kg} \\ 150 \text{ kg} \end{array} \right\} \text{Rockwell}$ $\left. \begin{array}{l} 15 \text{ kg} \\ 30 \text{ kg} \\ 45 \text{ kg} \end{array} \right\} \text{Superficial Rockwell}$	

^aFor the hardness formulas given, P (the applied load) is in kg and D , d , d_1 , and l are all in mm.



Slip phenomenon

Metals with FCC or BCC crystal structures have a relatively large number of slip systems (at least 12). These metals are quite ductile because extensive plastic deformation is normally possible along the various systems. Conversely, HCP metals, having few active slip systems, are normally quite brittle.

<i>Metals</i>	<i>Slip Plane</i>	<i>Slip Direction</i>	<i>Number of Slip Systems</i>
<i>Face-Centered Cubic</i>			
Cu, Al, Ni, Ag, Au	{111}	$\langle 110 \rangle$	12
<i>Body-Centered Cubic</i>			
α -Fe, W, Mo	{110}	$\langle 111 \rangle$	12
α -Fe, W	{211}	$\langle 111 \rangle$	12
α -Fe, K	{321}	$\langle 111 \rangle$	24
<i>Hexagonal Close-Packed</i>			
Cd, Zn, Mg, Ti, Be	{0001}	$\langle 11\bar{2}0 \rangle$	3
Ti, Mg, Zr	{10 $\bar{1}$ 0}	$\langle 11\bar{2}0 \rangle$	3
Ti, Mg	{10 $\bar{1}$ 1}	$\langle 11\bar{2}0 \rangle$	6



Om Namah shivay...

Thank you !!!

